25900265 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBOTA06 - PRINCIPLES OF MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or two Sentences

Answer any **10** questions

Each question carries **2** marks

1. Explain why algae are considered primary producers in aquatic ecosystems. [K] / [CO1]
2. Name the two photosystems associated with light reaction of photosynthesis. [K] / [CO2]
3. Write one advantage of using biofertilizers over chemical fertilizers. [U] / [CO4]
4. Why is bacterial morphology important in microbiology? [K] / [CO1]
5. Define fermentation. [K] / [CO4]
6. What is the composition of Bacterial cell membrane? [K] / [CO1]
7. What is transformation in bacteria? [U] / [CO2]
8. Why is Gram’s staining important in microbiology? [U] / [CO3]
9. State one limitation of SCP use as food. [U] / [CO4]
10. Name an example of a phage that follows a lysogenic cycle. [U] / [CO2]
11. What are anaerobic bacteria? [U] / [CO2]

12. What is a cloning vector? [U] / [CO2]

[2x10 = 20]

Part B

Short Answer

Answer any **8** questions

Each question carries **5** marks

13. Describe the major applications of amylases in the food and textile industries. [K] / [CO4]

14. Outline the process of conjugation in bacteria. [K] / [CO2]

15. List the essential elements and growth factors required for bacterial growth, and briefly describe their role. [K] / [CO2]

16. Describe the structure of Gram negative bacterial cell wall. [K] / [CO1]

17. Write short notes on endospores. [K] / [CO2]

18. Outline a microbial growth curve. [U] / [CO3]

19. Write a brief note on culture media used in microbiology. [U] / [CO3]

20. Write a brief note on bioremediation. [U] / [CO4]

21. Draw a labeled diagram of a typical prokaryotic cell and a eukaryotic animal cell. Highlight at least three key differences in structure. [K] / [CO1]

22. Differentiate lytic cycle from lysogenic cycle of viral replication. [U] / [CO2]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Write an essay on Glycolysis. [U] / [CO2]

24. Write an essay on the application of microbiology. [An] / [CO4]

[10x1 = 10]